

clausekeeper — ISO 9001:2015 traceability over a SuperDocs session

What & for whom

Quality managers inherit a quality manual plus five procedures that never quite prove coverage. Clausekeeper keeps those six documents in one SuperDocs multi-document session, maps every ISO 9001:2015 clause to the sections that cover it with provenance (`chunk_id` , heading path, verbatim quote), re-checks when procedures are edited, and raises orphaned clauses as gaps naming the exact edit (`change_id` , instruction, job, time).

Measured

Every number below comes from a committed run output or test assertion — never estimated. Paths are relative to `use-cases/sambaths/clausekeeper` .

Measure	Result	Trace
Keyless test suite	46 passed in ~10 s	<code>EVIDENCE.md:10</code> and <code>docs/media/transcripts/clausekeeper-b6-pytest.txt</code>
Full demo cycle	9 billable ops (≤10 gate)	<code>docs/media/transcripts/clausekeeper-b1-gap.txt</code> (init 0 + link 6 + edit 1 + show) + <code>docs/media/transcripts/clausekeeper-b3-pack.txt</code> (pack 1 + exports free) = 9; <code>EVIDENCE.md</code> usage ledger
Pack generation + exports	1 op for generation, DOCX + PDF via pre-signed downloads, <code>X-Export-Warnings</code> checked	<code>docs/media/transcripts/clausekeeper-b3-pack.txt</code> (<code>ops_spent:1</code> , <code>warnings: ["font Calibri substituted on page 2"]</code>) and <code>tests/fixtures/pack_flow.json</code>
Discover lane free-read	12/27 linkable clauses shortlisted at 0 ops	<code>docs/media/transcripts/clausekeeper-b4-discover.txt</code> (<code>local match resolved 12/27 ... (0 ops)</code>) and <code>tests/test_discover.py</code>
Discover lane batched search	one turn, +1 op capped, 15 remaining clauses via search	<code>docs/media/transcripts/clausekeeper-b4-discover.txt</code> (<code>[search] charged 1 op, search resolved 15/15</code>) and <code>tests/fixtures/discover_link.json</code>
Discover plan preview	0 ops	<code>docs/media/transcripts/clausekeeper-b4-discover.txt</code> (<code>DISCOVER PLAN (preview mode – zero billable ops)</code> and projected search spend: 1 op)
Discover weak fallback	WARN fallback at 1 op , no fabricated candidates	<code>docs/media/transcripts/clausekeeper-b4-weak.txt</code> (<code>WARN: search returned usable candidates for only 0 of 15 unresolved clauses, WARN: discover could not shortlist 15 clauses – continuing with standard mapping</code>) and <code>tests/fixtures/discover_weak.json</code>
Gap lands on approve	<code>chg_42</code> at 2026-08-22T14:02:00Z , instruction Delete the disposition section , job <code>job_edit_01</code>	<code>docs/media/transcripts/clausekeeper-b1-gap.txt</code> and <code>tests/fixtures/edit_delete.json</code>
Rename survival	<code>durable_document_id</code> matrix intact after NM-PRO-04 → NM-PRO-04-v2	<code>docs/media/transcripts/clausekeeper-b2-rename.txt</code> and <code>tests/test_guards.py::test_rename_keeps_links_and_refreshes_display_name</code>
Seeded gaps detected	2/2 (9.2 Internal audit, 7.4 Communication)	<code>docs/media/transcripts/clausekeeper-b1-gap.txt</code> (seed SEED-H1 clause 9.2: DETECTED , seed SEED-H2 clause 7.4: DETECTED) and <code>corpus/northgate/expected-gaps.yaml</code>

Why trade-offs

- SQLite matrix keyed on `durable_document_id` → renames survive display-name churn → single-writer, no concurrent editors; acceptable for a QMS control room
- One compact verification turn per document (`1 op/doc`) → costs stay predictable → slower gap surfacing within a cycle vs per-clause calls
- Free structural reads + at-most-one batched search for Discover → honors op budget → search precision limited by heading phrasing; advisory only
- Local gap-landing on `chunk_id` delete/edit → synchronous attribution (`chg_42` narrative) → only covers sections that were previously linked; new uncovered clauses rely on `show`
- Template upload once, pack generation one turn, pre-signed `D0CX` + `PDF` exports free via `POST /v1/downloads` with `X-Export-Warnings` decoded → branded output without per-page ops → export warnings are advisory, not blocking
- Fixture-replay tests (`tests/fixtures/*.json`) → fully keyless, deterministic, stranger-runable → live lane still hand-verified, not simulated

Honest limitations

Keyless tests mean the live path was verified by hand, not in CI; readiness statements cover the ISO 9001:2015 clause *structure* only — not an audit, they grant no standing and no external body has reviewed them; verification quality depends on the model tier behind the session; gaps can be false positives until a human reviews them at the approval flow.

Live-test note (2026-08-23, from `scratch/issues/assets/0006-live-uat-discover.md`): `free-read` works at `0 ops`, batched search exceeded `300 s` and fabricated candidate on seeded-gap `9.2` — worse than plain mapping on honesty. The free-read lane completed live at `0 ops` for five local clauses; the single batched search turn exceeded SuperDocs documented `verification_turn` max-wait (`300 s`), the CLI aborted before mapping and no shortlist rows persisted, and the model returned a plausible-looking section for seeded-gap clause `9.2` where no covering document exists by design. The later rerun completed within the envelope at `0 ops` but returned zero usable candidates with honest `WARN` fallback — shortlists remain advisory only, nothing feeds the mapping turns automatically, and every search-sourced row should be treated as unverified until a human checks it. See also `docs/media/transcripts/clausekeeper-b4-weak.txt` for the honest `WARN` path on film.

What's next

In order, each with what gets dropped: **(1)** surgical edit contract — outline-anchor edits with numbering-matches-placement validation, refuse placeholder tables where the document already contains the field (drop auto-styling normalization); **(2)** job observability — live progress events, structured errors everywhere, resumable sessions (drop synchronous blocking endpoints for large docs); **(3)** Google Docs add-on first — read via host API, edit via four calls, write back range-by-range (drop import-format breadth for a quarter); **(4)** a public document-editing benchmark — SWE-bench for documents, doubling as the regression harness (drop model-provider optionality breadth).

Run it yourself

`use-cases/sambaths/clausekeeper` on `superdocs-builds` PR #137 (`clausekeeper-core`), commit `d20cb18` .

```
` bash pip install -r requirements.txt python -m pytest -q # → 46 passed in ~10 s (fixture-replay, no API key) — see
docs/media/transcripts/clausekeeper-b6-pytest.txt python -m clausekeeper --fixture tests/fixtures/init_link.json init
python -m clausekeeper --fixture tests/fixtures/init_link.json link CK_AUTO_APPROVE=1 python -m clausekeeper --
fixture tests/fixtures/edit_delete.json edit NM-PRO-04 --instruction "Delete the disposition section" python -m
clausekeeper --db /tmp/ck-b1.db show # gap chg_42 with durable_document_id — see
docs/media/transcripts/clausekeeper-b1-gap.txt python -m clausekeeper --fixture tests/fixtures/pack_flow.json --db
/tmp/ck-pack.db pack --template assets/northgate-letterhead.docx --out /tmp/ck-exports # 1 op + X-Export-Warnings
— see docs/media/transcripts/clausekeeper-b3-pack.txt python -m clausekeeper --fixture
```

tests/fixtures/discover_link.json link --discover # free-read 12/27 at 0 ops + one batched search 15 rows at 1 op — see docs/media/transcripts/clausekeeper-b4-discover.txt python -m clausekeeper --fixture tests/fixtures/discover_link.json link --plan --discover # 0 ops python -m clausekeeper --fixture tests/fixtures/discover_weak.json link --discover # WARN fallback — see docs/media/transcripts/clausekeeper-b4-weak.txt `

This video is fully agent-generated with synthetic narration (Kokoro TTS `am_eric`, open-source) — no human on camera.